LAKSHMI SHIBU RAJ

SOFTWARE DEVELOPMENT ENGINEER (BACKEND & DISTRIBUTED SYSTEMS)
Brooklyn, NY | (347) 671-9120 | shiburajlakshmi@gmail.com | linkedin.com/in/lakshmi-shibu-raj-1149661b2/ | GitHub

PROFESSIONAL OVERVIEW

Results-driven **Software Development Engineer** with experience building scalable backend systems, distributed microservices, cloud-native applications, and full-stack platforms across banking, enterprise software, and cloud environments. Proven expertise in **Java**, **Spring Boot**, **REST APIs**, **AWS**, **Docker**, **Kubernetes**, and **microservices architecture**. Demonstrated success in improving system reliability, reducing API latency, automating operational processes, and delivering high-availability production systems. Experienced in designing **backend services**, **cloud deployments**, **CI/CD pipelines**, and scalable software solutions with measurable impact on performance, throughput, and operational efficiency. Strong foundation in algorithms, software engineering, system design, and **cloud-native** application development.

PROFESSIONAL EXPERIENCE

Software Engineering Intern | SIMPLIAN SOFTWARE SOLUTIONS, Diamond Bar, California **Jun 2025 – Jul 2025**

- Developed scalable Java Spring Boot solution enabling dynamic control of 20+ menu items with real-time automated updates
- Designed efficient backend operations, significantly reducing administrative dependency from five personnel to just one
- Built a self-service management interface cutting menu update time from 10 minutes to 2 minutes, boosting operational efficiency
- Partnered with product and engineering teams to deliver maintainable, production-ready enterprise software solutions
- Streamlined processes and improved system usability, significantly enhancing overall business productivity and efficiency

Software Development Engineer / Backend Developer | CITICORP SERVICES, India **Jul 2023 – Aug 2024**

- Designed a Java Spring Boot service eliminating 10+ daily failures, preventing critical pod crashes across enterprise applications
- Refactored core API framework across 15+ distributed services, reducing recurring incidents by 20+ critical production cases
- Tuned JVM and Tomcat configurations, cutting average API response latency by ~18% under peak production workload conditions
- Automated enterprise employee data collection with Python/Excel, saving 200+ staff hours weekly in Chief of Staff division
- Enhanced deployment reliability, sustaining consistent 99.98% uptime in dynamic Agile cross-functional enterprise environments

Software Engineering Intern | ROYAL BANK OF SCOTLAND, India **May 2022 – Jul 2022**

- Developed an interactive dashboard analyzing 15+ KPIs, improving reporting efficiency by 50% across business functions
- Automated workflows and pipelines for 5+ marketing teams, driving 30% engagement growth and 12% retention improvement
- Embedded real-time analytical insights into CRM, reducing manual reporting effort by 40% and accelerating decision cycles
- Strengthened BI capabilities through scalable data-driven solutions and automation initiatives supporting enterprise operations
- Delivered actionable insights enabling marketing, sales, and operations teams to improve efficiency and customer outcomes

PROJECT EXPERIENCE

NEIGHBORLY – SCALABLE COMMUNITY PLATFORM *(Link)*

- Architected React/Django community platform delivering 30+ REST APIs, 20+ reusable components, JWT auth, and Mapbox geolocation
- Deployed AWS EC2/S3/CloudFront with Docker CI/CD pipelines, reducing deployment effort 40% and achieving 100% test pass rate

COLLABORATIVE CODE EDITOR *(Link)*

- Built collaborative platform leveraging serverless AWS architecture with Lambda, API Gateway, DynamoDB, and S3 integrations
- Implemented scalable backend services enabling real-time collaboration, improving development efficiency and system reliability

TO-DO APPLICATION *(Link)*

- Developed containerized microservices architecture on AWS EKS/Kubernetes, integrating MongoDB persistence and Prometheus monitoring
- Orchestrated Docker releases with CI/CD pipelines, reducing deployment effort 40% while ensuring scalable, production-ready reliability

DINING CONCIERGE CHATBOT *(Link)*

- Built AWS serverless restaurant recommendation chatbot with Lex, Lambda, API Gateway, DynamoDB, OpenSearch, and S3 integrations
- Implemented event-driven workflows and email delivery via SES, reducing manual effort and enhancing real-time user interaction

EDUCATION

Master of Science in Computer Science | New York University, New York | **GPA:** 3.6 **May 2026**
Bachelor of Technology in Information Technology | National Institute of Technology, India | **GPA:** 3.14 **Apr 2023**

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, S3, Lambda, CloudFront, API Gateway, DynamoDB, OpenSearch, SQS, SES, CloudWatch, EKS), Kubernetes, Docker, Linux | **Backend Engineering:** Java Spring Boot, Microservices Architecture, REST APIs, Distributed Systems, Node.js, Django, Tomcat, Maven | **DevOps & CI/CD:** CI/CD Pipelines, Git, Containerization, Deployment Automation, Infrastructure Management | **AI/ML Technologies:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas | **Databases:** MongoDB, MySQL, Redis, DynamoDB, NoSQL Databases | **Programming Languages:** Java, Python, C++, SQL, JavaScript | **Frontend Development:** React.js, JavaScript, HTML5, CSS3

CORE COMPETENCIES

Backend Software Development | Distributed Systems Engineering | Microservices Architecture | REST API Development | System Design & Scalability | Cloud-Native Application Development | AWS Architecture & Deployment | Kubernetes Orchestration | Docker Containerization | CI/CD Pipeline Automation | Performance Optimization | High Availability Systems | Database Design & Optimization | Full Stack Development | Agile Software Development | Production Support & Reliability Engineering | Infrastructure Automation | Software Architecture & Technical Problem Solving